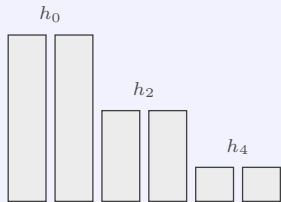
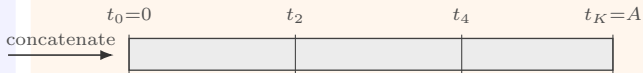


### A PER-UNIT TABLES



unit tables:  $K \approx 3.6 \cdot 10^4$  columns, heights  $h_y$  from 32 rows to millions

### B DENSE PACKING AND STACKING



dense vector  $\mathbf{q} = \mathbf{c}_0 \parallel \cdots \parallel \mathbf{c}_{K-1}$ , length  $A$  (trace area), padded to  $2^m$ ;  
 $\text{col}(i)$  and  $\text{row}(i) = i - t_{\text{col}(i)}$  locate position  $i$

cut into  $2^{m-\ell}$  stripes of  $2^\ell$  entries



$\cdots$  committed once

32 stripes = one Merkle leaf per coset row

### C REDUCTION OF A COLUMN CLAIM

